

Lingjun Liu

lliu39@ncsu.edu | lingjunliu.github.io | linkedin | github.com/lingjunliu

EDUCATION

- North Carolina State University (NCSU)**, Raleigh, NC, USA 2024 – 2028 (Expected)
Ph.D. in Computer Science
Adviser: Prof. Tim Menzies
- Korea Advanced Institute of Science and Technology (KAIST)**, Daejeon, South Korea 2019 – 2021
M.S. in Computer Science
Adviser: Prof. Doo-Hwan Bae
Exchange Program in Computer Science (during B.S.) 2018 – 2019
- National Tsing Hua University**, Hsinchu, Taiwan 2014 – 2019
B.S. in Computer Science
Note: Five-year degree including one-year exchange at KAIST

PUBLICATIONS

- **Learning Compiler Fuzzing Mutators from Historical Bugs** [preprint]
Lingjun Liu, Feiran Qin, Owolabi Legunsen, Marcelo d'Amorim
International Mining Software Repositories Conference (MSR), 2026
- **Development of Reliability Measurement Method and Tool for Nuclear Power Plant Safety Software** [TKIPS]
Lingjun Liu, Wooyoung Choi, Eunkyoun Jee, Duksan Ryu
The Transactions of the Korea Information Processing Society, 2024
– Extended version of the KCSE 2024 paper
- **A Reliability Evaluation Tool for Nuclear Power Plant Safety Software**
Lingjun Liu, Wooyoung Choi, Eunkyoun Jee, Duksan Ryu
Korea Conference on Software Engineering (KCSE), 2024
Best Short Paper Award 🏆
- **Search-based Test Case Selection for PLC Systems using Functional Block Diagram Programs** [ISSRE]
Miriam Ugarte Querejeta, Eunkyoun Jee, Lingjun Liu, Pablo Valle, Aitor Arrieta, Miren Illarramendi Rezabal
International Symposium on Software Reliability Engineering, 2023
- **MuFBDTester: A mutation-based test sequence generator for FBD programs implementing nuclear power plant software** [STVR]
Lingjun Liu, Eunkyoun Jee, Doo-Hwan Bae
Software Testing, Verification and Reliability, 2022
- **An Empirical Study of Reliability Analysis for Platooning System-of-Systems** [QRS-C]
Sangwon Hyun, Lingjun Liu, Hansu Kim, Esther Cho, Doo-Hwan Bae
International Conference on Software Quality, Reliability and Security Companion, 2021
- **Attack-driven Test Case Generation Approach using Model-checking Technique for Collaborating Systems** [EnCyCriS]
Zelalem Mihret, Lingjun Liu
International Workshop on Engineering and Cybersecurity of Critical Systems, 2021.
- **Platooning LEGOs: An Open Physical Exemplar for Engineering Self-Adaptive Cyber-Physical Systems-of-Systems** [SEAMS]
Yong-Jun Shin, Lingjun Liu, Sangwon Hyun, Doo-Hwan Bae
International Symposium on Software Engineering for Adaptive and Self-Managing Systems, 2021
Best Artifact Award 🏆
- **MuGenFBD: Automated Mutant Generator for Function Block Diagram Program** [KTSDE]
Lingjun Liu, Eunkyoun Jee, Doo-Hwan Bae
KIPS Transactions on Software and Data Engineering, 2021
– Extended version of the KCSE 2020 paper
- **A Systematic Translation from PAT-based Counterexamples to Viable Test Cases**
Zelalem Mihret, Lingjun Liu, Eunkyoun Jee, Doo-Hwan Bae
Korea Conference on Software Engineering (KCSE), 2021
- **Analysis of coupling effect hypothesis for function block diagram programs**
Lingjun Liu, Eunkyoun Jee, Doo-Hwan Bae
Korea Software Congress (KSC), 2020
- **Automated mutant generation for function block diagram programs**
Lingjun Liu, Eunkyoun Jee, Doo-Hwan Bae
Korea Conference on Software Engineering (KCSE), 2020
Outstanding Short Paper Award 🏆

ACADEMIC EXPERIENCE

North Carolina State University
Graduate Research Assistant

Raleigh, NC, USA
2024 – Present

- (Research Project) Mutational Fuzzing for C Compilers
 - Built program mutation-based fuzzing framework using agentic AI (Gemini) to automatically generate bug history-driven mutators for C compilers
 - Discovered 60 bugs across GCC and Clang

KAIST
Full-time Researcher

Daejeon, South Korea
2023 – 2024 | 2021 – 2022

- (Research Project) Reliability Estimation of Nuclear PLC Software
 - Developed reliability measurement prototype using Bayesian Belief Networks and statistical testing to enable regulatory organizations to evaluate nuclear PLC software reliability

KAIST
Graduate Research Assistant

Daejeon, South Korea
2018 – 2021

- (Research Project) Mutation-Based Test Generation for FBD Programs
 - Developed automated test generator for nuclear safety software using mutation operators and Yices SMT solver
 - Modeled FBD programs and mutations as SMT constraints to enable systematic test case generation
 - Maximized fault detection effectiveness through mutation-based testing techniques
- (Research Project) Runtime Verification of System-of-Systems
 - Designed runtime verification property patterns and scopes
 - Developed verification checker for Mass Casualty Incident-Response (MCI-R) systems to ensure safety properties during operation
- (Research Project) Security Testing of Air Traffic Control Systems
 - Modeled attack scenarios on air traffic control systems using JADE multi-agent framework
 - Implemented test case generation via model checking to systematically verify system behaviors under attacks
- (Research Project) Failure Pattern Analysis for Transportation Systems
 - Applied sequence pattern mining and time-series clustering to detect interaction failures

INDUSTRY EXPERIENCE

Suresoft Technologies Inc.
Associate Research Engineer

Seongnam, South Korea
2022 – 2023

- Static Analysis Tool Development
 - Developed C++ rule checkers based on AUTOSAR C++14 coding guidelines to improve automotive software quality
 - Implemented backend services for license validation using Java Spring framework
 - Managed software packaging and conducted regression testing to ensure reliability

FELLOWSHIPS

- 2024 University Graduate Fellowship at NCSU – Top graduate student fellowship
- 2019 KAIST Scholarship – Merit-based full support

PRESENTATIONS

MuFBDTester: A mutation-based test sequence generator for FBD programs implementing nuclear power plant software – [Link](#)

- ISSRE – Journal First, Conference Second (J1C2) Track 2023
- KCSE – Invited talk of excellent international conference/journal papers 2023